

Estimation of the exponential growth rate of an epidemic

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ABSTRACT

Accurate estimation of the initial growth rate of an epidemic is critical for assessing transmissibility and guiding early interventions. Standard regression-based methods, such as negative binomial regression, often rely on independence assumptions that may underestimate uncertainty. We propose a hidden Markov model (HMM) framework that explicitly accounts for the unobserved infectious population. Using data generated from a stochastic linear SEIR model, we compare the performance of different methods for estimating the exponential growth rate. Our results show that HMMs yield more robust and reliable estimates than negative binomial regression, with improved coverage probabilities of the 95 % credible intervals. In particular, modelling the infectious population with a negative binomial distribution (with a constant probability parameter) or a binomial distribution provides more accurate inference than the other methods considered. Moreover, this framework is extended beyond the exponential phase by combining it with a logistic model to account for the slowing down of the exponential growth due to the depleting of the susceptible population. We apply this extended method to data on the early stage of COVID-19 pandemic in Africa and Ontario, Canada to estimate the initial growth rate, and we demonstrate that the HMM-logistic framework improves stability and reliability of estimates.

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1. Introduction

In the early stages of an epidemic outbreak, the number of infections generally grows exponentially. The exponential growth rate is a fundamental measure in infectious disease epidemiology, describing how quickly the number of cases increases. It serves as a direct indicator of the speed of disease transmission and provides a basis for estimating the basic reproduction number, $\mathcal{R}_0$ (Chowell, Shim, et al., 2006; Lipsitch et al., 2003; Mills et al., 2004; Roberts & Heesterbeek, 2007). Furthermore, the growth rate plays a key role in evaluating the effectiveness of control measures, since a reduction in its value indicates the success of interventions such as non-pharmaceutical interventions (NPIs), contact tracing, or vaccination (Chowell, Ammon, et al., 2006; Elliott et al., 2021; Ma et al., 2014). Accurately estimating this rate enables public health officials to optimize response strategies, efficiently allocate medical resources, and anticipate the future burden of disease, ensuring a timely and effective outbreak response.

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Statistical methods have been widely used to estimate the exponential growth rate of an outbreak, including the least squares method (Björck, 1990), maximum likelihood estimation (MLE) (Bolker, 2008; Myung, 2003), and Bayesian Markov chain Monte Carlo (MCMC) (Cauchemez et al., 2004). A common approach is to apply a linear regression model using the least squares method, which involves taking the logarithm of observed case counts and fitting a straight line to the data over time (Chowell et al., 2003; Mills et al., 2004; Vynnycky et al., 2007). This method is simple and computationally efficient, but assumes constant variance and normally distributed errors, which may not hold in real-world outbreak data.

Poisson regression models are another commonly used approach for parameter estimation via the maximum likelihood method (Coxe et al., 2009; Hutchinson & Holtman, 2005). In this approach, the number of new cases is assumed to follow a Poisson distribution, and infection events are modeled as having exponentially distributed waiting times. However, Poisson regression assumes that the variance is equal to the mean, which may not be realistic in the presence of overdispersion, in which case counts exhibit greater variability than expected under a Poisson distribution. To address overdispersion and underdispersion in count data, Bayesian Conway–Maxwell–Poisson regression models have been proposed (Huang & Kim, 2021). These models use Bayesian inference via the Metropolis–Hastings algorithm and have been evaluated through data analysis and simulations. Negative binomial regression models are often used to fit overdispersed data, such as the case counts and the secondary transmission cases (Bi et al., 2020; Endo et al., 2020; Laxminarayan et al., 2020). However, these methods are dependent on data collected during the early stages of an epidemic, in which the exponential growth phase is short and available data are limited (Ma et al., 2014).

On the other hand, Poisson regression models, as well as negative binomial regression models, typically assume that observations are independent. This means that the occurrence of one event does not influence the occurrence of another, and each observation is considered to be unrelated to others. However, in real-world scenarios, especially during the early stages of an epidemic, this assumption may not hold. Observations are not independent but instead correlated over time, which can affect the accuracy of the estimation results. For example, in the stochastic linear SEIR model discussed in (Wang et al., 2025), we assumed that daily new cases are obtained through the detection of infectious individuals. Therefore, the observations of daily new cases are not independent. This dependence arises because new cases on a given day are generated by infections that occurred in previous days, meaning today's incidence is directly linked to the size of the infectious population yesterday. The hidden Markov model (HMM) is another approach commonly used to describe systems that evolve over time in a way that is partially observable (Fort et al., 2015; Mor et al., 2021). The model assumes that the system is in one of a set of hidden states at any given time, but these states cannot be directly observed. Instead, only the observable outputs or observations, which are probabilistically related to the hidden states, are observed. Since epidemic processes are usually only partially observed through routine surveillance data, HMMs are often applied in epidemiology (Cooper & Lipsitch, 2004; Green & Richardson, 2002; Le Strat & Carrat, 1999).

Both MLE and MCMC methods require a clear understanding of the distribution of case counts to establish appropriate likelihood functions for parameter estimation. In (Wang et al., 2025), we showed that when time t is very small, the binomial distribution can approximate the probability distribution of daily new cases. However, as the outbreak enters the exponential growth phase, the data quickly exhibit overdispersion, making the negative binomial distribution a better fit for case counts. Building on this finding, Section 2 introduces two models for estimating the exponential growth rate: the negative binomial regression model and the HMM, and evaluates the performance of different likelihood functions in terms of estimation accuracy. Section 3 applies our proposed methods to the COVID-19 outbreak in Africa and Ontario, Canada to estimate the exponential growth rate. Concluding remarks are given in Section 4.

2. Methods

2.1. Negative binomial regression model

In (Wang et al., 2025), we found that the number of new cases is approximately binomially (near the very beginning) or negative binomially distributed (after a short period from emergence) in the exponential growth stage of an epidemic. By the time the spread is detected in the population, the number of cases is typically in an exponential growth phase, and the binomially distributed phase has passed. Therefore, under the assumption that observations are independent, it is reasonable to model the distribution of cases as a negative binomial distribution.

In the negative binomial regression model, the daily new cases, $C(t)$, follow a negative binomial distribution:

$$C(t) \sim NB(r, p),$$

where r is the size parameter, and p is the probability parameter.

The expected value of $C(t)$ is modeled as an exponential growth function:

$$\mathbb{E}[C(t)] = c_0 e^{\lambda t},$$

where c_0 is the expected number of $C(0)$ and λ is the growth rate. Unlike a Poisson distribution, a negative binomial distribution cannot be uniquely determined by its mean, because the mean $\mathbb{E}[C(t)] = \frac{r(1-p)}{p}$. Thus, either p or r needs to be estimated with model parameters c_0 and λ , while the other may then be inferred from the mean. To address this

identifiability structure, we consider two commonly used parameterization approaches (Aryuyuen & Tonggumnead, 2022; Hilbe, 2011). These parameterizations correspond to different assumptions about the dispersion of epidemic count data, and each provides a distinct interpretation of the underlying variability in daily case counts.

Method 1: Negative binomial likelihood with a constant probability parameter p . This parameterization assumes that the probability parameter p remains constant over time, implying that the proportion of variance relative to the mean stays approximately stable as cases grow. The alternative size parameter then becomes time-dependent:

$$r_t = \mathbb{E}[C(t)] \frac{p}{1-p} = \frac{c_0 e^{\lambda t} p}{1-p},$$

reflecting the level of dispersion at time t .

This approach is appropriate when overdispersion increases proportionally with the mean. Using the PMF of the negative binomial distribution, the likelihood function is:

$$\mathcal{L}_1[C(t)|p, c_0, \lambda] = \prod_{t=1}^T \left(\frac{C(t) + r_t - 1}{C(t)} \right) p^{r_t} (1-p)^{C(t)}. \quad (1)$$

Method 2: Negative binomial likelihood with a constant size parameter r . Here, the size parameter r is assumed constant, meaning that the absolute amount of overdispersion does not grow with the magnitude of daily cases. Instead, the probability parameter becomes time-dependent:

$$p_t = \frac{r}{r + \mathbb{E}[C(t)]} = \frac{r}{r + c_0 e^{\lambda t}}.$$

This representation is widely used when variability is dominated by persistent reporting heterogeneity or clustering effects that do not scale with epidemic growth. The corresponding likelihood function is:

$$\mathcal{L}_2[C(t)|r, c_0, \lambda] = \prod_{t=1}^T \left(\frac{C(t) + r - 1}{C(t)} \right) p_t^r (1-p_t)^{C(t)}. \quad (2)$$

2.1.1. Accuracy of exponential growth rate estimation

To compare the accuracy of these two methods in estimating the exponential growth rate λ , we conducted a simulation study. Given that many diseases, such as influenza and COVID-19, have a latent period, we employed a stochastic linear Susceptible-Exposed-Infectious-Removed (SEIR) model (Wang et al., 2025). In this model, the transmission rate is denoted by β , the latent period by $1/\sigma$, and infectious individuals are detected and move to compartment R at a rate γ . Assuming that $S \approx N$ during the initial exponential growth phase, we generated 100 datasets of $C(t)$ over a time range of 0–90 days using fixed parameter values motivated by COVID-19: transmission rate $\beta = 0.4$ (Ge et al., 2021), mean latent period of 5 days ($\sigma = 0.2$) (Li et al., 2020), and detection rate $\gamma = 0.2$ (Rong et al., 2020). We generated 100 independently simulated datasets using these parameter values. The simulation codes are provided as the supplementary material.

We estimated λ within a Bayesian framework, employing the *nimble* package with MCMC sampling for posterior inference. For each simulated dataset, the model was run with a single MCMC chain consisting of 200,000 iterations, of which the first 100,000 were discarded as burn-in. No thinning was applied. Initial values were set to the prior means for all parameters. The prior distributions used for all parameters are summarized in Table 1. Convergence was assessed by visual inspection of trace plots. Posterior summaries (mean and 95 % credible intervals) were then extracted from the remaining 100,000 post-burn-in samples for each parameter and each dataset.

The goodness-of-fit is evaluated using the mean squared relative error (MSRE), which is defined as

$$\text{MSRE} = \frac{1}{T} \sum_{t=1}^T \left(\frac{C(t) - \hat{C}(t)}{\hat{C}(t)} \right)^2, \text{ where } \hat{C}(t) = \gamma I_0 e^{\lambda t},$$

Table 1

Prior distributions used for parameter estimation in the Bayesian framework.

Parameter	Definition	Prior distribution
p	Probability parameter	Normal($\mu = 0.6, \sigma = 0.3$)
r	Negative binomial size parameter	Normal($\mu = 10, \sigma = 4$)
I_0	Initial number of infectious individuals	Normal($\mu = 10, \sigma = 4$)
c_0	Expected number of new cases at $t = 0$	Normal($\mu = 5, \sigma = 1.5$)
λ	Exponential growth rate	Normal($\mu = 0.08, \sigma = 0.04$)

and T is the length of the fitting window. It quantifies relative deviations between observed and fitted counts, which is particularly useful for epidemic data where the daily cases are not independently identically distributed (Fissler & Ziegel, 2019; Gneiting, 2011).

For each simulated dataset, we evaluated model performance across different fitting windows, where each window corresponds to the time period of observed data used for estimating the exponential growth rate. Specifically, we considered nine fitting windows ranging from $[0, 10]$ to $[0, 90]$ in increments of 10 days. Fig. 1 displays how three key diagnostics depend on different fitting windows: the point estimates of λ , the width of the corresponding 95 % credible intervals, and the MSRE, with the average (lines) and the middle 95 % quantiles across 100 simulated datasets). The left panel shows the point estimates (MCMC sample mean). For short fitting windows (less than 50 days), both methods tend to underestimate λ , with Method 1 exhibiting slightly larger bias. As the fitting window increases and more data are used, the point estimates from both methods approach the true value of λ (used to generate the data). The middle panel reports the width of the 95 % credible intervals. This width decreases steadily with larger fitting windows, reflecting improved precision. The right panel compares the MSRE. As the fitting window length increases and more data are included, the average MSRE decreases for both methods, indicating improved model fit. Overall, both parameterizations show consistent trends, though their performance differs for short data windows. It is interesting that Method 2 shows a slight advantage, even though theoretical studies reveal that the case distributions is consistent with Method 1.

When estimating the growth rate λ , a key question is whether our credible intervals accurately capture the value of λ used to generate the data (true λ), or merely reflect the specific epidemic realization we observe. Real epidemics are just one outcome of a random process. Different outbreaks can have different observed growth rates, even if they follow the same transmission rules. The credible intervals should properly quantify this uncertainty. To assess whether our credible intervals are reliable, we examine their coverage probability, which measures how often the credible intervals contain the true λ over many realizations. A good method should have a coverage probability that matches the nominal confidence level. The coverage probability is then estimated by calculating the proportion of the 95 % credible intervals that include the true λ across the 100 estimates.

Table 2 presents the estimated coverage probabilities for λ using two different parameterization methods. Both methods exhibit coverage probabilities below the nominal 95 % level, with Method 2 generally achieving higher coverage than Method 1. Coverage probability reflects both bias in the point estimates and the width of the credible intervals. As shown in Fig. 1, both methods tend to underestimate λ for short and intermediate fitting windows, with Method 1 showing larger bias. Method 2 produces point estimates closer to the value of λ used to generate the data, explaining its relatively higher coverage despite similar interval widths. For short fitting windows (e.g., fewer than 10 days), coverage can appear larger than expected (even 100 %) because the credible intervals are very wide due to limited data, not because the estimates are more accurate. For intermediate window lengths (around 50 days), under-coverage occurs: the estimator remains biased (underestimating λ) while intervals are too narrow, leading to coverage below 95 %. For longer windows, the intervals narrow further, but persistent bias prevents coverage from reaching the nominal level.

Despite exploring two parameterization methods within the negative binomial regression framework, our results indicate that this approach does not reliably estimate the exponential growth rate λ . Both methods suffer from substantial bias for short and intermediate fitting windows, and even in longer windows, the credible intervals remain too narrow relative to the estimator's bias, leading to persistent undercoverage. A key reason for this poor performance is that the negative binomial regression model treats the daily case counts $C(t)$ as independent. In reality, however, $C(t)$ depends on the hidden number of infectious individuals $I(t)$ at time t , and $I(t)$ evolves. This induces serial dependence in the observed data

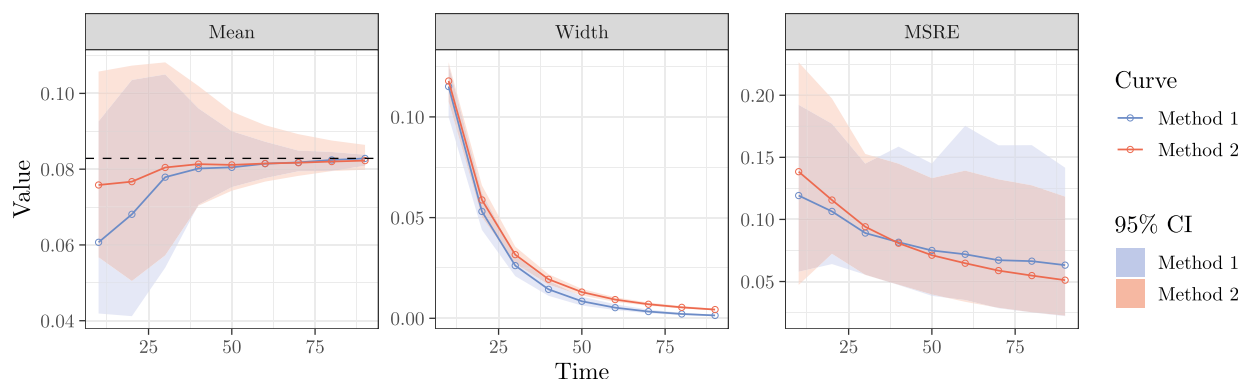


Fig. 1. Comparison of the point estimate (MCMC sample mean) of the exponential growth rate λ (the left panel), the width of the 95 % credible interval (CI) of λ (the middle panel), and the mean squared relative error (MSRE) of daily new cases (the right panel) across different fitting windows. For each fitting window, 100 simulated datasets were fitted, producing 100 estimates. The lines represent the average across the 100 fits, and the shaded area represents the middle 95 % quantiles of the 100 fits. Blue represents Method 1 (negative binomial likelihood with a constant probability parameter p), and red represents Method 2 (negative binomial likelihood with a constant size parameter r). In the left panel, the horizontal dashed line indicates the value $\lambda \approx 0.084$ used to generate the data, computed using the fixed parameters $\beta = 0.4$ and $\sigma = \gamma = 0.2$.

Table 2Estimated coverage probability of growth rate λ for Methods 1 and 2 over various fitting windows.

Time Period	Method 1	Method 2
[0, 90]	61 %	68 %
[0, 80]	60 %	68 %
[0, 70]	52 %	76 %
[0, 60]	60 %	76 %
[0, 50]	48 %	65 %
[0, 40]	61 %	68 %
[0, 30]	60 %	76 %
[0, 20]	68 %	85 %
[0, 10]	100 %	100 %

that the regression model cannot capture. As a result, parameter estimates derived under the independence assumption fail to reflect the true underlying dynamics.

To address these limitations, we turn to a hidden Markov model (HMM) framework, which allows us to treat $I(t)$ as a hidden process governing the observed case counts $C(t)$ and explicitly incorporate the temporal dependence between them.

2.2. The hidden Markov model

The number of daily new cases, $C(t)$, depends on whether the infectious individuals $I(t)$ are detected at a rate γ , and $I(t)$ depends on $I(t-1)$. At the same time, $C(t-1)$ also depends on $I(t-1)$. Thus $C(t)$ and $C(t-1)$ are not independent. Theoretically, $I(t)$ and $C(t)$ have the same exponential growth rate provided that each infectious individual has the same probability of detection. Thus it might be easier to estimate the growth rate from $I(t)$. However, obtaining data for $I(t)$ is difficult, as it is not directly observed. Therefore, we consider $I(t)$ as a hidden state (an unobserved variable), and use the HMM to capture the dependencies in the data. The HMM allows us to model the hidden state $I(t)$ while using observable data $C(t)$ to infer it. We can estimate the unobserved state transitions over time and account for the dependencies between the hidden and observable states.

Here, we model the expected value of $I(t)$ as exponential growth:

$$\mathbb{E}[I(t)|I(t-1)] = I(t-1)e^{\lambda}.$$

This is a discrete-time model with unit time steps. Given $I(t)$, the observed cases are assumed to be binomial distributed (Joe, 2016)

$$C(t)|I(t) \sim \text{Bin}(I(t), \gamma).$$

To establish the likelihood function, we need to specify the distribution for $I(t)$. When we assume that $I(t)$ follows a negative binomial distribution:

$$I(t) \sim \text{NB}(r, p),$$

we can establish the likelihood function in two different ways as follows.

Method 3: HMM with negative binomial likelihood and a constant probability parameter p . Like Method 1, we assume p is a constant and estimate it with other model parameters. The size parameter r can be written in terms of p and $\mathbb{E}[I(t)]$:

$$r_t = \mathbb{E}[I(t)] \frac{p}{1-p} = \frac{I(t-1)e^{\lambda}p}{1-p}.$$

The joint likelihood function (Madsen & Royle, 2023) is given by

$$\begin{aligned} \mathcal{L}_3(C|\lambda, p, I_0, \gamma) &= \prod_{t=1}^T [\text{NB}(I(t)|r_t, p) \cdot \text{Bin}(C(t)|I(t), \gamma)] \\ &= \prod_{t=1}^T \left(\frac{I(t) + r_t - 1}{I(t)} \right) p^{r_t} (1-p)^{I(t)} \left(\frac{I(t)}{C(t)} \right) \gamma^{C(t)} (1-\gamma)^{I(t)-C(t)}. \end{aligned} \quad (3)$$

Method 4: HMM with negative binomial likelihood and a constant size parameter r . Like Method 2, we assume that r is a constant and estimate it with other model parameters. We express the probability parameter p in terms of r and $\mathbb{E}[I(t)]$:

$$p_t = \frac{r}{r + \mathbb{E}[I(t)]} = \frac{r}{r + I(t-1)e^\lambda}.$$

Then, the likelihood function is:

$$\begin{aligned} \mathcal{L}_4(C|\lambda, r, I_0, \gamma) &= \prod_{t=1}^T [NB(I(t)|r, p_t) \cdot \text{Bin}(C(t)|I(t), \gamma)] \\ &= \prod_{t=1}^T \left(\frac{I(t) + r - 1}{I(t)} p_t^r (1 - p_t)^{I(t)} \left(\frac{I(t)}{C(t)} \right) \gamma^{C(t)} (1 - \gamma)^{I(t) - C(t)} \right). \end{aligned} \quad (4)$$

2.2.1. Additional HMM methods

In (Wang et al., 2025), we found that when t is large, the distribution of $I(t)$ can be approximated by a negative binomial distribution. However, when time t is small, the variance of $I(t)$ is less than its mean and $I(t)$ can be approximated by a binomial distribution. For this HMM, we assume that the time step is 1, so the distribution of $I(t)$ may follow a binomial distribution.

Method 5: HMM with a binomial likelihood and a constant probability parameter P . We assume that $I(t)$ follows a binomial distribution,

$$I(t) \sim \text{Bin}(N_t, P),$$

and the size parameter N_t is proportional to $I(t-1)$, i.e., $N_t = aI(t-1)$, where a is a constant and $a > 1$. Then, the probability is

$$P = \frac{\mathbb{E}[I(t)]}{N_t} = \frac{I(t-1)e^\lambda}{aI(t-1)} = \frac{e^\lambda}{a}.$$

The likelihood function can be written as:

$$\begin{aligned} \mathcal{L}_5(C|\lambda, a, I_0, \gamma) &= \prod_{t=1}^T [\text{Bin}(I(t)|N_t, P) \cdot \text{Bin}(C(t)|I(t), \gamma)] \\ &= \left(\frac{N_t}{I(t)} \right)^{I(t)} P^{I(t)} (1 - P)^{N_t - I(t)} \left(\frac{I(t)}{C(t)} \right) \gamma^{C(t)} (1 - \gamma)^{I(t) - C(t)}. \end{aligned} \quad (5)$$

The negative binomial distribution is suitable for handling overdispersed data, while the binomial distribution is appropriate for underdispersed data. For completeness, we also consider the Poisson distribution, which has equal mean and variance.

Method 6: HMM with the Poisson likelihood. If $I(t)$ follows a Poisson distribution,

$$I(t) \sim \text{Poisson}(\mathbb{E}[I(t)]),$$

where $\mathbb{E}[I(t)] = I(t-1)e^\lambda$. Using the PMF of Poisson distribution, the likelihood function is given by

$$\begin{aligned} \mathcal{L}_6(C|\lambda, I_0, \gamma) &= \prod_{t=1}^T [\text{Poisson}(I(t)|\mathbb{E}[I(t)]) \cdot \text{Bin}(C(t)|I(t), \gamma)] \\ &= \frac{\mathbb{E}[I(t)]^{I(t)} e^{-\mathbb{E}[I(t)]}}{I(t)!} \left(\frac{I(t)}{C(t)} \right) \gamma^{C(t)} (1 - \gamma)^{I(t) - C(t)}. \end{aligned} \quad (6)$$

2.2.2. Comparison of accuracy

To compare the accuracy of these four likelihood functions for estimating the growth rate λ , we fit the data of $C(t)$ using a Bayesian HMM with the *nimble* package. In these HMMs, the parameters we aim to estimate include the exponential growth rate λ , the initial number of infectious individuals I_0 , and the negative binomial or binomial distribution parameters (p or r depending on the method). The prior distributions for all parameters are summarized in Table 1. For each simulated dataset, we ran a single MCMC chain for 200,000 iterations, with the first 20,000 iterations discarded as burn-in.

We observed that when the detection rate γ is estimated jointly with λ and the distribution parameters, the MCMC fails to converge. As an example, Fig. 2 shows the trace plots for Method 5 on a representative dataset. Here, λ converges, but the other parameters do not, revealing a strong dependency between γ and I_0 : an increase in γ is compensated by a decrease in I_0 , resulting in nearly identical trajectories of the observed cases $C(t)$. This indicates that the primary non-identifiability occurs between γ and I_0 . The resulting posterior ridge leads to poor convergence for γ , I_0 , and p . Fixing γ eliminates this non-identifiability and allows reliable estimation of λ . Although informative priors on γ could partially constrain the

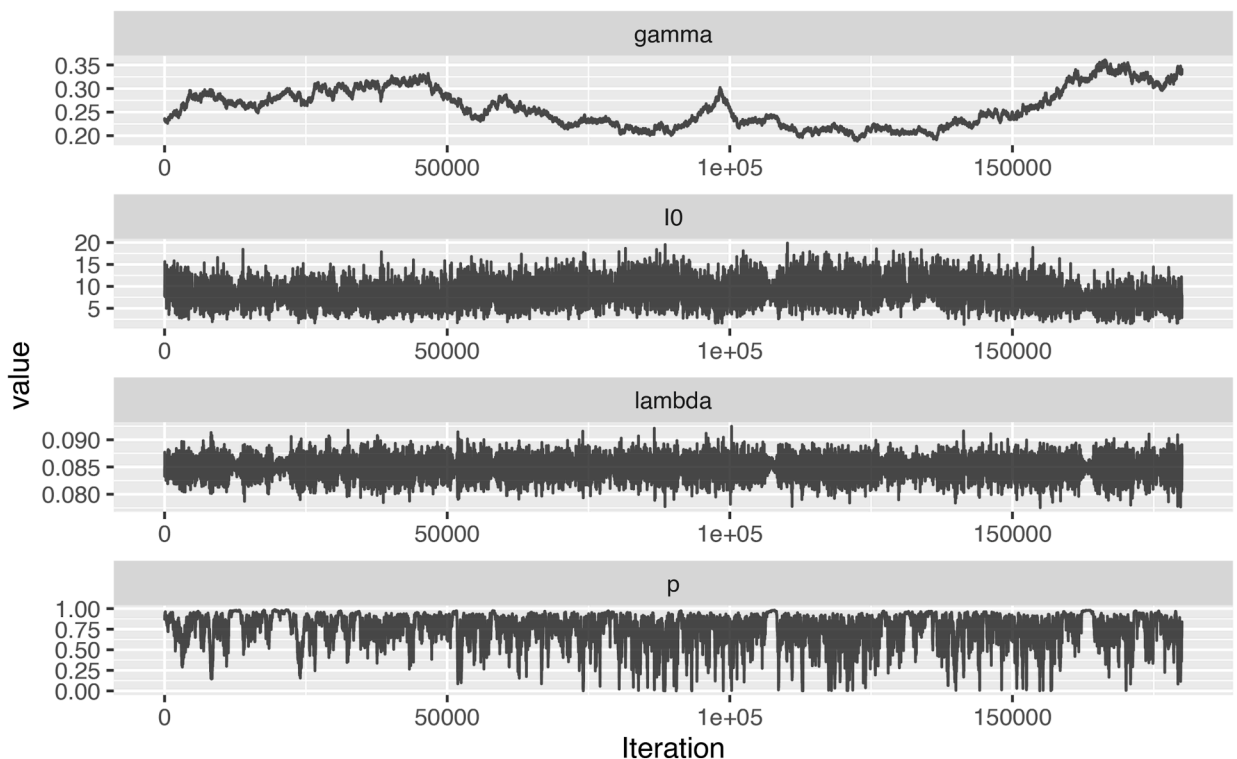


Fig. 2. Trace plots for Method 5. The detection rate γ is estimated jointly with the growth rate λ , the initial number of infectious individuals I_0 , and the probability parameter p . Here, the chain converges for λ but not for γ , I_0 and p .

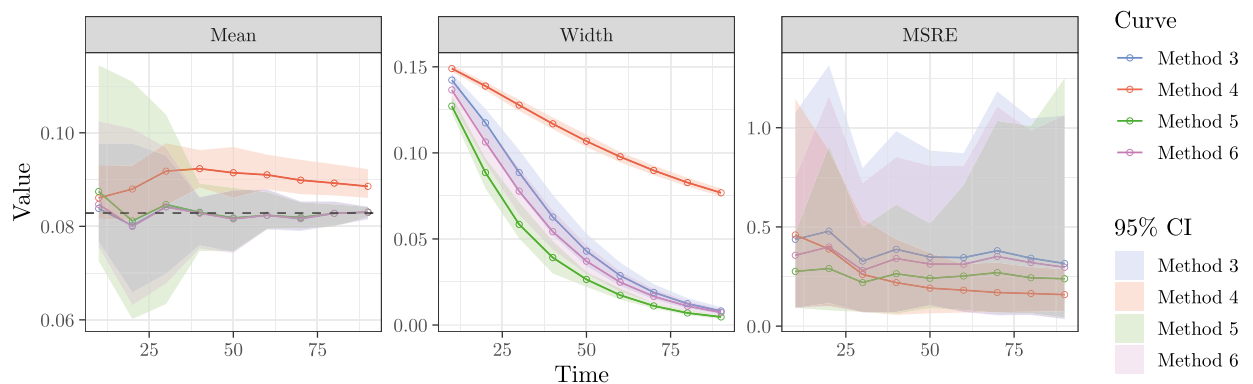


Fig. 3. Dependence of the estimated exponential growth rate λ on the fitting window. For each fitting window, 100 independently simulated datasets were fitted, producing 100 point estimates (the left panel) and 100 corresponding 95 % CIs (the middle panel presents their width). The mean of these estimates are represented by lines, and the middle 95 % quantiles are represented by the shaded areas. The right panel reports the MSRE obtained from the 100 fits. Blue represents Method 3 (negative binomial with a constant p), red represents Method 4 (negative binomial with a constant r), green represents Method 5 (binomial distribution), and pink represents Method 6 (Poisson distribution). For the left panel, the horizontal dashed line represents the λ used to generate the data.

parameter space, fixing γ provides a simpler and more robust solution for the dataset considered in this paper. Therefore, in later simulations, we set $\gamma = 0.2$ and estimate λ and the remaining parameters using their posterior means. The R code used to implement these methods is included as supplementary material.

Fig. 3 illustrates the performance of four different methods (Methods 3, 4, 5, and 6) in estimating the exponential growth rate λ across various fitting windows. As in Fig. 1, 100 datasets are randomly generated with the same parameter values. The left panel shows the point estimates of λ . The horizontal dashed line represents the true value of λ . Method 4 (negative binomial with constant r) exhibits the largest bias and deviation from the true λ . In contrast, Methods 3, 5, and 6 follow a similar trend and approach the true λ rapidly as the fitting window extends beyond 50 days, demonstrating reduced bias

with increased data. The middle panel presents the width of the 95 % credible intervals. As expected, uncertainty generally decreases as the fitting window extends and more data are included. Method 4 consistently produces the widest intervals, indicating higher uncertainty in its parameter estimates compared to the other three methods. The right panel shows the MSRE across fitting windows. All methods maintain relatively stable and low average MSRE values throughout the fitting windows. Method 4 yields the best MSRE, especially with longer fitting windows. This figure shows that a method with a better goodness-of-fit may not yield more precise estimates of the parameter values.

The poor performance of Method 4 suggests that holding the size parameter r fixed imposes a rigid variance-to-mean relationship ($\text{Var}(I) = \mathbb{E}[I] + \mathbb{E}[I]^2 / r$) that is unsuitable for capturing the dynamic overdispersion structure of growing epidemic count data. Following the results in (Wang et al., 2025), the size parameter increases exponentially over time. Fixing r leads to substantial bias. In contrast, the performance of the negative binomial model with a constant probability parameter p (Method 3), is comparable to the simpler, less overdispersed-sensitive Poisson (Method 6) and binomial (Method 5) distributions. This success suggests that, within this HMM structure, allowing the effective “dispersion” to be dynamically linked to the changing mean via a constant probability p offers the necessary flexibility to accurately model the evolving relationship between the count and its variability during the growth phase.

This comparison is conducted using a transmission rate β and a latent period $1/\sigma$ when generating the data of $C(t)$. When the removal rate γ is fixed, there are infinitely many combinations of β and σ that yield the same exponential growth rate. Consequently, our method cannot identify β and σ individually. The dependence of the results on the combinations of β and σ is addressed below.

2.2.3. Dependence of the estimation results on β and σ

Based on the formula for λ derived in (Wang et al., 2025),

$$\lambda = \frac{-(\sigma + \gamma) + \sqrt{(\sigma - \gamma)^2 + 4\sigma\beta}}{2}, \quad (7)$$

we fix $\lambda \approx 0.084$, $\gamma = 0.2$, and vary σ from 0.2 to 1 in increments of 0.2, adjusting β accordingly. The corresponding parameter values are shown in Table 3. For each set of parameters, we generated 100 datasets of $C(t)$. As shown previously in Fig. 3, Method 4 (negative binomial with constant r) exhibits substantial bias in estimating λ . Therefore, our comparison here focuses on the remaining three methods. Fig. 4 summarizes how the estimation results depend on the underlying parameters (σ, β). The left panel shows that all three methods produce point estimates of λ that remain close to the true λ for all parameter settings, with only mild fluctuations as σ increases. The middle panel reveals clear differences in uncertainty across methods: Methods 3 and 6 show decreasing credible interval widths as σ increases, while Method 5 maintains nearly constant widths across all parameter values. Among the three, Method 5 consistently yields the narrowest intervals. The right panel shows that the average MSRE values increase slightly with increasing σ for all three methods, and all methods follow the same general trend.

We compare the estimated coverage probability of the 95 % credible intervals for the three methods in Table 3. Method 3 and Method 5 exhibit high coverage for all parameter settings. However, Method 6, which assumes a Poisson likelihood, provides lower coverage except when we have precise knowledge of the latent period rate $1/\sigma$.

2.2.4. Dependence of the estimation results on γ

In the previous simulation studies, we fixed the detection rate γ because jointly estimating γ with λ and the distribution parameters led to clear nonidentifiability: γ and the remaining parameters failed to converge, whereas fixing γ yielded stable and accurate inference for λ . To evaluate the robustness of this modelling choice, we conducted a sensitivity analysis to examine how the estimation results vary when different fixed values of γ are used. In this analysis, γ was varied over a range of values smaller than the transmission rate $\beta = 0.4$. This choice ensures that $\beta > \gamma$, so that the basic reproduction number $\mathcal{R}_0 = \beta/\gamma$ remains greater than 1, consistent with epidemic growth in our simulation setting.

We focus on Methods 3 and 5 in this analysis because earlier simulation results demonstrated that these two methods consistently provide reliable inference: they produce accurate point estimates of λ , narrower credible intervals, and high coverage probabilities of the true λ . For a sequence of fixed γ values, we fitted 100 simulated datasets for each value while estimating λ and the remaining parameters. The results are summarized in Fig. 5. The mean estimates of λ produced by both methods remain close to the true value across all choices of γ , indicating that the point estimator is robust to the assumed

Table 3

Dependence of the estimated coverage probability of the growth rate λ on the parameters σ and β , where β is calculated from equation (7).

σ	β	Method 3	Method 5	Method 6
0.2	0.4	100 %	100 %	100 %
0.4	0.341	100 %	100 %	68 %
0.6	0.322	100 %	99 %	68 %
0.8	0.312	100 %	98 %	68 %
1	0.306	100 %	99 %	60 %

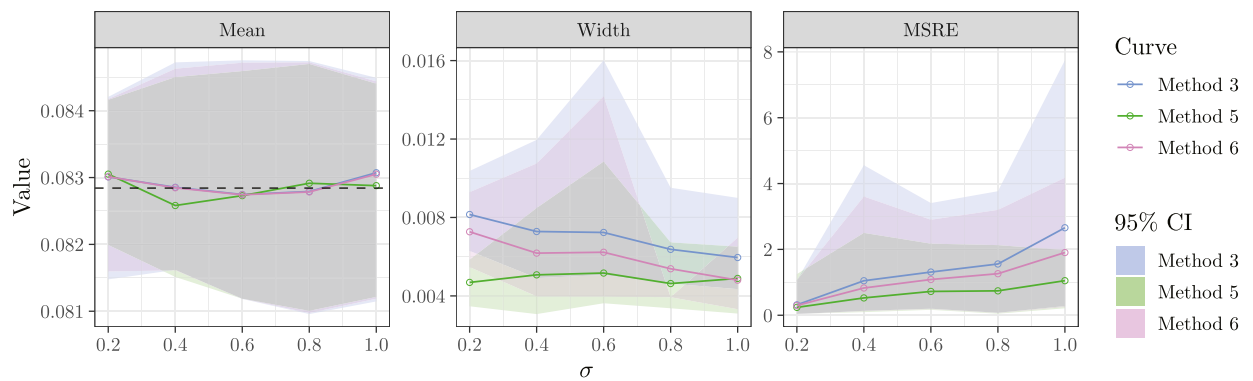


Fig. 4. Dependence of the estimated exponential growth rate λ on model parameters. For each parameter setting, 100 simulated datasets were fitted, producing 100 point estimates of λ and 100 corresponding 95 % CIs. The left panel presents the mean of the 100 point estimates of λ , together with their empirical 95 % confidence intervals. The middle panel shows the mean width of the 100 corresponding 95 % CIs. The right panel reports the average MSRE obtained from the 100 fits. Blue represents Method 3 (negative binomial with a constant p), green represents Method 5 (binomial distribution), and pink represents Method 6 (Poisson distribution). For the left panel, the horizontal dashed line represents the λ used to generate the data.

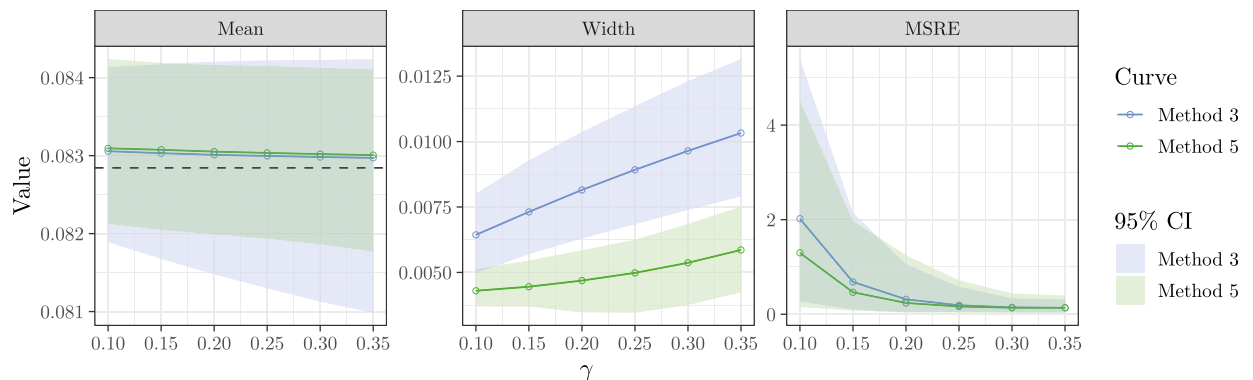


Fig. 5. Dependence of the estimated exponential growth rate λ on the detection rate γ . For each value of γ , 100 simulated datasets were fitted, where γ was treated as a fixed parameter during inference while λ and the remaining parameters were estimated. The lines represent the average, and the shaded areas represent the middle 95 % quantiles of the 100 fits. The left panel presents the point estimates of λ . The middle panel shows the width of the corresponding 95 % CIs. The right panel reports the average MSRE. Blue represents Method 3 (negative binomial with a constant p) and green represents Method 5 (binomial distribution). For the left panel, the horizontal dashed line represents the λ used to generate the data.

detection rate. The width of the 95 % credible intervals increases with larger γ . Method 5 yields more stable and consistently narrower intervals. The average MSRE values decrease as γ increases, suggesting an improvement in model fit when a larger fraction of infections is detected.

These results also suggest that if some prior information about γ is available, such as a plausible range informed by testing capacity or epidemiological studies, this information can be incorporated to stabilize the estimates further. By constraining γ to a realistic interval, the estimation of λ and other parameters becomes more reliable, reducing the impact of nonidentifiability while still allowing for uncertainty in γ .

Method 4 gives the largest bias in λ , while Method 6 gives low coverage probabilities for the 95 % credible intervals for imprecise latent period values. Methods 3 and 5 both yield small biases in λ and are stable for imprecise latent periods and detection rates. Method 5 yields a slightly narrower credible interval and a better goodness-of-fit.

3. Application to the COVID-19 pandemic in Africa

Musa et al. (Musa et al., 2020) analyzed the early stage of the COVID-19 pandemic in Africa. They estimated the initial growth rate using daily new case data from March 1 to March 19, 2020. The epidemic curve was modeled with exponential growth, following (Ma, 2020), and the generation interval (GI) was assumed to follow a Gamma distribution with a mean of 4.7 days and a standard deviation of 2.9 days. Their model, fitted using a Poisson likelihood framework, yielded a growth rate of 0.22 per day (95 % CI: 0.20–0.24). The fitted epidemic curve was then projected and compared with observed data up to April 13, 2020 (Fig. 6).

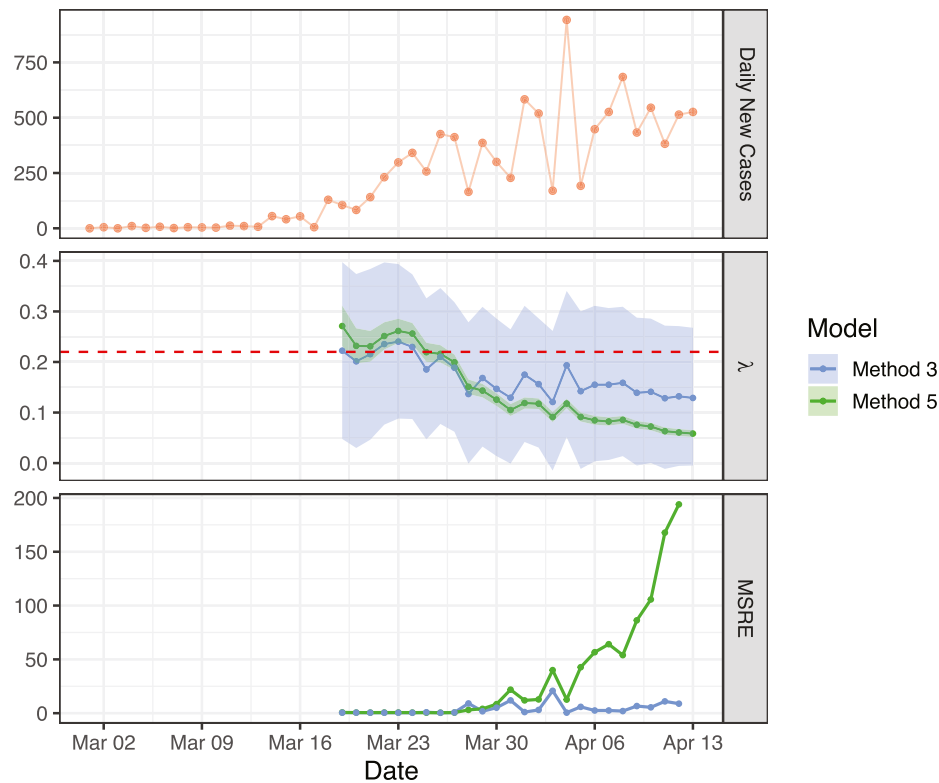


Fig. 6. Comparison of fitted exponential growth rates and their 95 % credible intervals as a function of the fitting window. The top panel shows the daily number of new cases in Africa from March 1 to April 13, 2020 (data source: World Health Organization (WHO) ([World Health Organization](#))). The middle panel displays the estimated growth rate λ as a function of the end date of the fitting window (with the start date fixed at March 1). The bottom panel shows the MSRE of the corresponding point estimates. Method 3 corresponds to the HMM with negative binomial likelihood and a constant p , while Method 5 corresponds to the HMM with binomial likelihood. Shaded regions represent the 95 % credible intervals. The red dashed line indicates the estimate of $\lambda = 0.22$ from ([Musa et al., 2020](#)).

In this section, we apply our proposed methods to the COVID-19 pandemic in Africa in 2020 to estimate the exponential growth rate during its early stage and compare the results with those reported in ([Musa et al., 2020](#)), using the same data provided by WHO (World Health Organization).

3.1. Estimation under a pure exponential growth phase

Through the simulations in the previous sections, we found that when the epidemic curve exhibits a pure exponential growth phase (with λ constant), the estimates of the growth rate λ obtained from Method 3 (HMM with negative binomial likelihood and constant p) and Method 5 (HMM with binomial likelihood) are closer to the true λ , and their 95 % credible intervals have higher coverage probabilities for the true value of λ . In these simulations, the epidemic data were generated using a linear SEIR model under the assumption that $S \approx N$. This assumption holds when the number of susceptible individuals is approximately equal to the total population. During the early exponential growth phase, the depletion of susceptibles is negligible, so the assumption is valid. As the epidemic progresses and S begins to decline, the accuracy of these two methods may be affected. Next, we apply these two methods to the data shown in [Fig. 6](#).

For all unknown parameters, we explicitly specify prior distributions and configure the MCMC settings to ensure stable posterior inference. Posterior distributions are estimated using MCMC implemented in nimble in R, with 200,000 iterations

Table 4
Prior distributions used for Bayesian HMMs applied to early COVID-19 data from Africa and Ontario, Canada (see subsection 3.3).

Parameter	Definition	Prior (Africa)	Prior (Ontario, Canada)
I_0	Initial infectious count	Uniform[1100]	Uniform[100,600]
λ (λ_0)	Initial growth rate	Normal($\mu = 0.22$, $\sigma = 0.1$)	Normal($\mu = 0.3$, $\sigma = 0.1$)
p	Probability parameter	Uniform[0,1]	Uniform[0,1]
K	Carrying capacity	Uniform[$\sum_t C(t)$, $4\sum_t C(t)$]	Uniform[$\sum_t C(t)$, $4\sum_t C(t)$]

per chain, a burn-in of 50,000, and a single chain, monitoring λ , p , and I_0 . The prior distributions for all parameters are summarized in Table 4. Posterior means and 95 % credible intervals are reported and visualized in Fig. 6 as shaded regions.

Fig. 6 compares the exponential growth rates fitted by Method 3 and Method 5, considering λ as a function of the end time of the fitting window. For each fitting window, we assume that the epidemic curve is purely exponential, so the corresponding λ is constant. The figure shows that for fitting windows shorter than 25 days, both methods yield results close to 0.22. As the window extends, the estimated value of λ decreases markedly, because the epidemic has moved beyond the exponential growth phase and the growth rate has begun to slow. Method 5 provides more stable estimates of λ , especially later in the time series; whereas, Method 3 consistently yields wider 95 % credible intervals, indicating greater uncertainty in its estimates.

In terms of model fit quality, the MSRE values for the two methods are almost identical and very small (below 1) when the fitting window ends before March 30, suggesting both methods fit the early exponential phase well. After this point, the MSRE of both methods begins to rise as the epidemic deviates from purely exponential growth. However, as the end time extends further into April, the MSRE for Method 5 increases sharply, while Method 3 shows only a modest increase. This indicates that Method 3 is less sensitive to deviations from exponential growth and maintains a more consistent fit over longer windows, whereas Method 5 deteriorates rapidly once the data no longer follow an exponential trend.

3.2. Estimation under a time-varying growth phase

In reality, the pure exponential growth phase of the epidemic curve is very short, as shown in Fig. 6, with the growth rate changing in response to control measures. To better capture the slowing of the epidemic as it approaches its final size, we model the cumulative cases using a logistic growth model (Ma, 2020). This phenomenological model accounts for the gradual deceleration of growth, providing more robust estimates of the epidemic's progression even when the growth rate is time-varying.

Here, the total number of cases $X(t)$ is modeled as:

$$\frac{d}{dt}X(t) = \lambda_0 X(t) \left(1 - \frac{X(t)}{K}\right), \quad (8)$$

where λ_0 is the initial growth rate and $K = \lim_{t \rightarrow \infty} X(t)$.

Since the daily new cases $C(t)$ correspond to the derivative of the cumulative cases, we have

$$C(t) = \frac{d}{dt}X(t).$$

Assuming that $C(t)$ initially grows approximately exponentially, we can write

$$C'(t) = \lambda(t)C(t),$$

which allows us to express the instantaneous growth rate as

$$\lambda(t) = \frac{C'(t)}{C(t)} = \frac{d^2X(t)/dt^2}{dX(t)/dt} = \lambda_0 - 2\lambda_0 \frac{X(t)}{K}.$$

Note that if we assume $C(t)$ corresponds to the discrete differences of the cumulative cases, i.e., $C(t) = X(t) - X(t-1)$, then using the discrete-time analogue of the logistic model (Merdan & Şişman, 2020), we can approximate

$$\lambda(t) = \lambda_0 - 2\lambda_0 \frac{X(t-1)}{K}. \quad (9)$$

We applied Method 5, incorporating a logistic model of cumulative cases, to estimate the initial growth rate λ_0 from the African COVID-19 data (World Health Organization). For the Logistic model, the carrying capacity K is assigned a uniform prior between $K_{\min} = \sum_t C(t)$ and $K_{\max} = 4\sum_t C(t)$. The other prior and MCMC settings are the same as those described for the Exponential model.

Fig. 7 shows the fitted values of λ_0 as a function of the end time of the fitting window. Compared with the Exponential model, which assumes purely exponential growth, the Logistic model accounts for epidemic saturation and provides more stable estimates over longer fitting windows. For shorter windows (ending before day 25), both models produce estimates fluctuating around the reference value of 0.22. As the fitting window extends beyond 25 days, the Exponential model estimates steadily decline below the reference value, whereas the logistic model mitigates this downward trend, stabilizing λ_0 around 0.16–0.18 by the end of the series. This difference reflects the deceleration of the epidemic as control measures take effect and the growth rate slows over time, suggesting that the logistic extension improves the stability and reliability of estimates when data extend beyond the purely exponential phase. The MSRE results further highlight this improvement. Across all fitting windows, the Logistic model maintains very small MSRE values, consistently below 1, indicating that it provides a good fit even as the epidemic departs from exponential growth. For reproducibility, the data and simulation code are provided in the supplementary files.

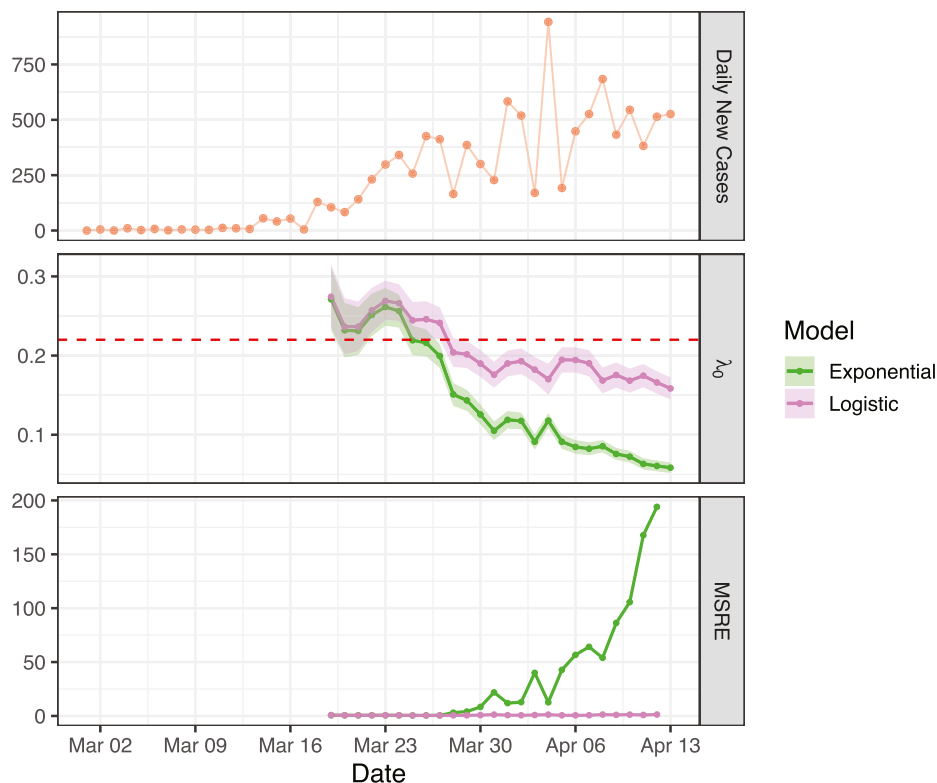


Fig. 7. Comparison of fitted initial growth rates and their 95 % credible intervals for Method 5 as a function of the end time of the fitting window. The top panel shows daily new cases in Africa from March 1 to April 13, 2020. The middle panel presents the estimated initial growth rate λ_0 as a function of the end date of the fitting window (with the start date fixed at March 1). The bottom panel shows the mean squared relative error (MSRE) of the corresponding model fits. The Exponential model corresponds to Method 5 (HMM with binomial likelihood), while the Logistic model combines Method 5 with a logistic formulation of cumulative cases to account for epidemic saturation. Shaded regions represent the 95 % credible intervals. The red dashed line indicates the estimated λ_0 of 0.22 from (Musa et al., 2020).

3.3. Validation using COVID-19 data from Ontario, Canada

To verify whether the results obtained from the African COVID-19 data are generalizable, we applied the same estimation methods to early pandemic data from Ontario, Canada, spanning March 13 to April 30, 2020. The prior distributions for all parameters are summarized in Table 4.

Fig. 8 shows the fitted exponential growth rates using Method 3 (HMM with negative binomial likelihood and constant p) and Method 5 (HMM with binomial likelihood) under the assumption of a pure exponential growth phase. Although both methods produce similar point estimates, Method 3 exhibits substantially wider 95 % credible intervals. Both methods remain stable for shorter fitting windows corresponding to the initial exponential phase, consistent with the patterns observed in the Africa analysis. Once the end time of the fitting window extends beyond April 6, however, the estimated growth rates from both methods begin to decline, reflecting the slowdown of the epidemic as control measures take effect.

The MSRE patterns in Fig. 8 further reinforce this interpretation. When the end time of the fitting window is earlier than April 6, both Method 3 and Method 5 yield very small MSRE values (all below 1), indicating an excellent fit during the early exponential phase. As the window extends beyond this point, the MSRE for both methods increases, coinciding with the observed decline in the estimated growth rates and signalling a clear departure from exponential growth. Consistent with the African dataset, Method 5 exhibits a noticeably larger MSRE than Method 3 for longer windows, suggesting that the binomial likelihood alone struggles to capture the increasing variability and systematic deviations from exponential behavior in the later stage of the epidemic.

Fig. 9 incorporates a logistic model of cumulative cases using Method 5 to account for epidemic saturation. Similar to the African dataset, the Exponential model estimates decline as the epidemic slows, whereas the Logistic model mitigates this downward trend and provides more stable estimates of the initial growth rate λ_0 over longer fitting windows. The MSRE results again highlight this improvement: the Logistic model maintains consistently small MSRE values (always below 1) across all fitting windows, demonstrating that it provides a good fit even when the epidemic curve begins to deviate from exponential growth. These results confirm that the patterns observed in Africa, namely, the need for a logistic extension to

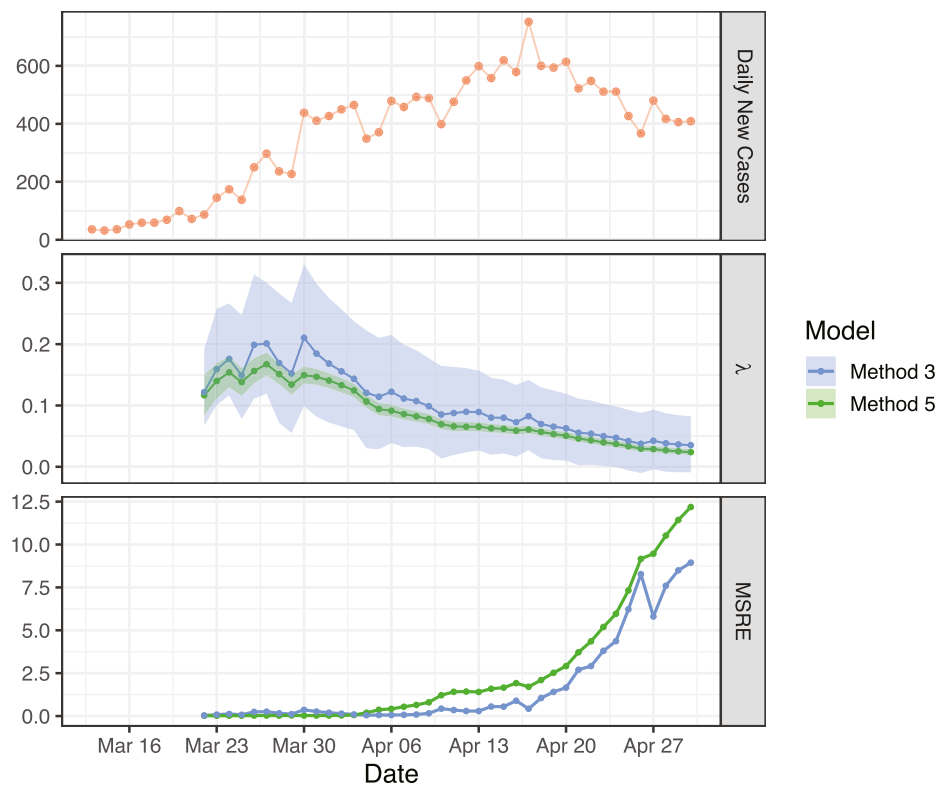


Fig. 8. Comparison of fitted exponential growth rates and their 95 % credible intervals as a function of the end time of the fitting window. The top panel shows the daily number of new cases in Ontario, Canada from March 13 to April 30, 2020 (data source: Government of Canada ([Government of Canada](#))). The middle panel presents the estimated growth rate λ as a function of the end date of the fitting window (with the start date fixed at March 1). The bottom panel shows the mean squared relative error (MSRE) of the corresponding model fits. Method 3 corresponds to the HMM with negative binomial likelihood and constant p , while Method 5 corresponds to the HMM with binomial likelihood. Shaded regions represent the 95 % credible intervals.

capture slowing growth, are also present in the Ontario data, demonstrating the robustness of the proposed methods across different regions.

4. Concluding remarks

We investigated statistical methods for estimating the exponential growth rate, λ , in the early stages of an outbreak. Accurate estimation of λ is vital for understanding disease transmission dynamics and guiding public health interventions, as it directly informs projections of epidemic growth and the potential impact of control measures. Based on the approximate distributions of infectious individuals $I(t)$ and daily new cases $C(t)$ derived in ([Wang et al., 2025](#)), we considered both the negative binomial regression model and hidden Markov models (HMMs). HMMs account for the hidden state $I(t)$, which, although unobservable, affects the observed daily new cases $C(t)$. Because time dependence is considered, the value of $I(t)$ depends on previous time steps.

We generated 100 sets of $C(t)$ data over a 90-day period using fixed parameter values from a stochastic linear SEIR model ([Wang et al., 2025](#)). We compared the results of different methods for estimating the initial growth rate λ with respect to the end time of the fitting window, keeping the start time fixed at 0. Our results highlight that HMMs accounting for the unobserved infectious population $I(t)$ generally yield 95 % credible intervals for λ with better coverage probabilities than the negative binomial regression model. In particular, within the HMM framework, methods that model $I(t)$ using a negative binomial distribution (with constant probability parameter p) or a binomial distribution provide the most accurate estimates of λ among the methods considered here.

We applied our proposed methods to the early stage of COVID-19 outbreak in Africa in 2020 ([World Health Organization](#)). During the exponential growth phase, both methods provided reasonable estimates of the growth rate. However, once the epidemic curve moved beyond the purely exponential phase and the growth rate declined, estimates of the initial growth rate became more sensitive. In particular, the HMM with a negative binomial distribution and constant p produced very wide 95 % credible intervals, introducing considerable uncertainty.

While our proposed method is suitable for the purely exponential growth phase of the epidemic curve, the method can be combined with a logistic model to better capture the slowing down of exponential growth near the peak of an epidemic.

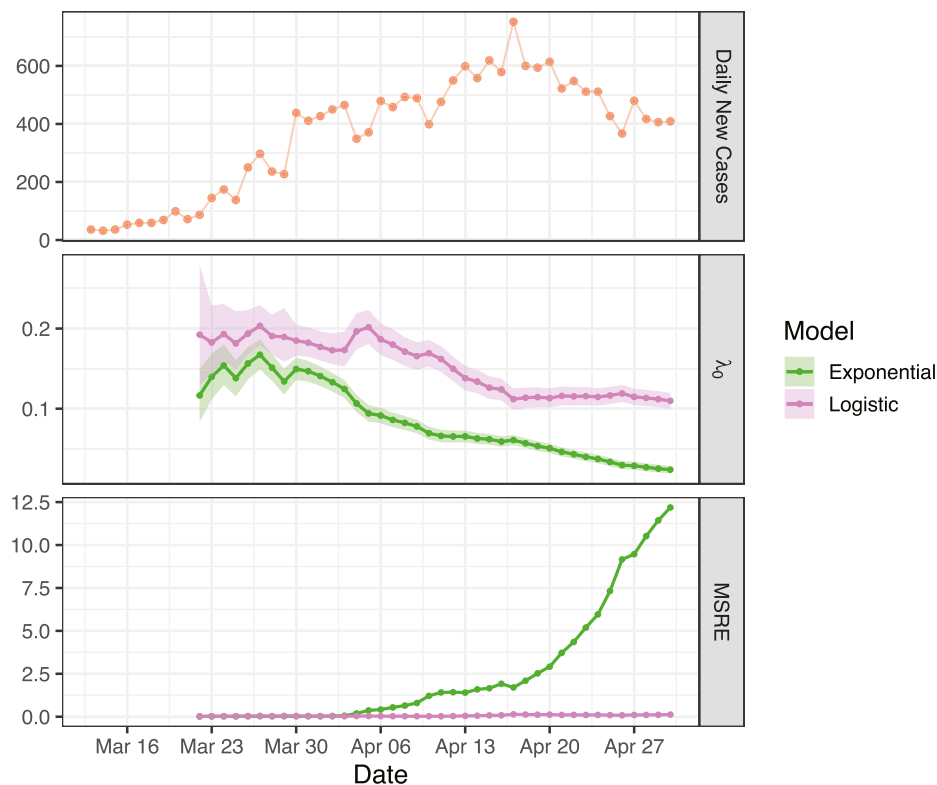


Fig. 9. Comparison of fitted initial growth rates and their 95 % credible intervals for Method 5 as a function of the end time of the fitting window. The top panel shows daily new cases in Ontario, Canada from March 13 to April 30, 2020. The middle displays shows the estimated initial growth rate λ_0 as a function of the end date of the fitting window (with the start date fixed at March 1). The bottom panel shows the mean squared relative error (MSRE) of the corresponding model fits. The Exponential model corresponds to Method 5 (HMM with negative binomial likelihood and constant p), while the Logistic model combines Method 5 with a logistic formulation of cumulative cases to account for epidemic saturation. Shaded regions represent the 95 % credible intervals.

We modeled the cumulative number of cases at time t using a logistic model and combined this with the HMM with a binomial distribution to estimate the initial growth rate. We applied this extended method to early COVID-19 data from Africa, where it provided more stable estimates of the initial growth rate even after case counts moved beyond the purely exponential phase. This approach also allowed us to derive a time-dependent growth rate that reflects its evolution over the course of the outbreak. To further verify the robustness of the method, we applied the same procedures to early COVID-19 data from Ontario, Canada, in 2020, and obtained qualitatively consistent results, confirming that the method performs reliably across different datasets.

Several recent methods aim to improve early-phase growth estimation by explicitly modelling the gradual deceleration of an epidemic. For example, Bartolomeo et al. (Bartolomeo et al., 2021) applied an exponential-decay model to weighted cumulative daily growth rates to capture the overall epidemic trend of COVID-19 in Italy, linking the exponential and slowdown phases. Their approach focuses primarily on describing the epidemic trajectory, while our HMM–logistic framework is specifically designed to estimate the initial exponential growth rate by incorporating the hidden infectious population and a mechanistic description of early epidemic dynamics.

After the peak, when cumulative case counts begin to decline, it remains an open question whether the case distribution should be modeled as negative binomial or binomial. An interesting future direction is to develop reliable methods for estimating the growth rate as it changes over longer periods. The growth rate may change discontinuously due to interventions or behavioural shifts. Our methods provide a reliable interval estimate for the exponential growth rate, which is crucial for detecting such change points. The framework could be extended by comparing models with and without a change to identify when a shift occurs.

CRedit authorship contribution statement

Manting Wang: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **P. van den Driessche:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Formal analysis. **Laura L.E. Cowen:** Writing – review & editing, Supervision, Funding acquisition. **Junling Ma:**

Writing – review & editing, Validation, Supervision, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Conflict of interest

The authors declare they have no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.idm.2025.12.020>.

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